Department of Electronics and Electrical Communication Engineering
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Embedded Systems Laboratory

EC39202

Experiment 2: Interfacing and Control of LEDs and Combinational Logic using 8051 Microcontroller



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Group 5

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1. Problem Statement

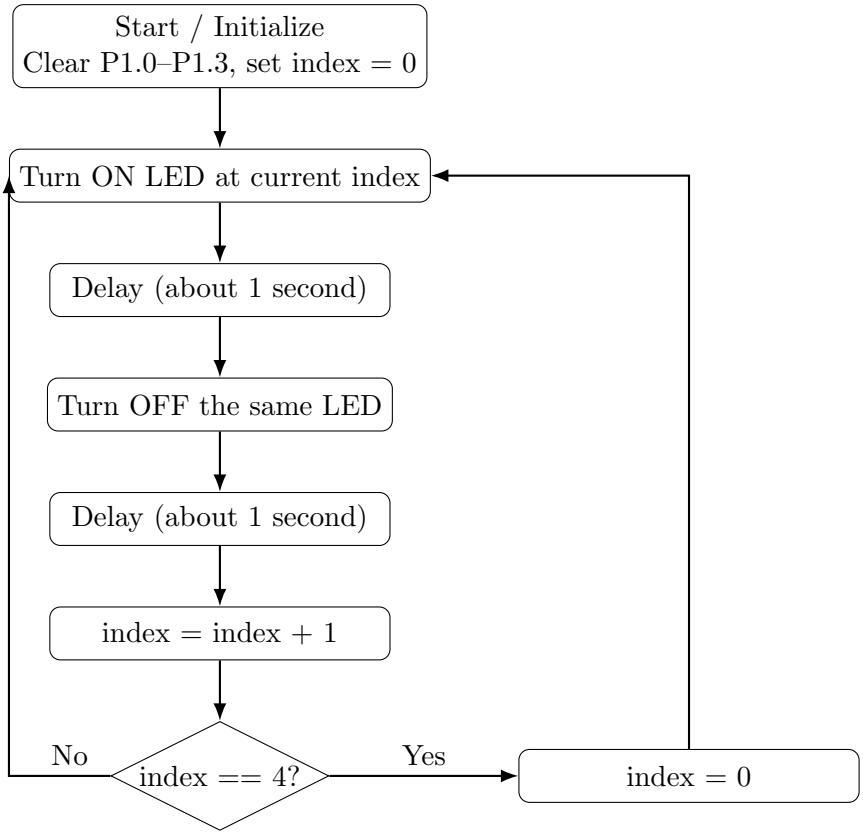
Objective:

1. Glow one LED and demonstrate basic LED control.
2. Blink four LEDs sequentially with ON and OFF duration of approximately 1 second each.
3. Implement a four-input combinational logic evaluator where:
   - Inputs are A (MSB), B, C, D (LSB).
   - Truth-table entries are stored in external memory locations 9000H to 900FH.
   - Four LEDs indicate input combination and one LED indicates function output.

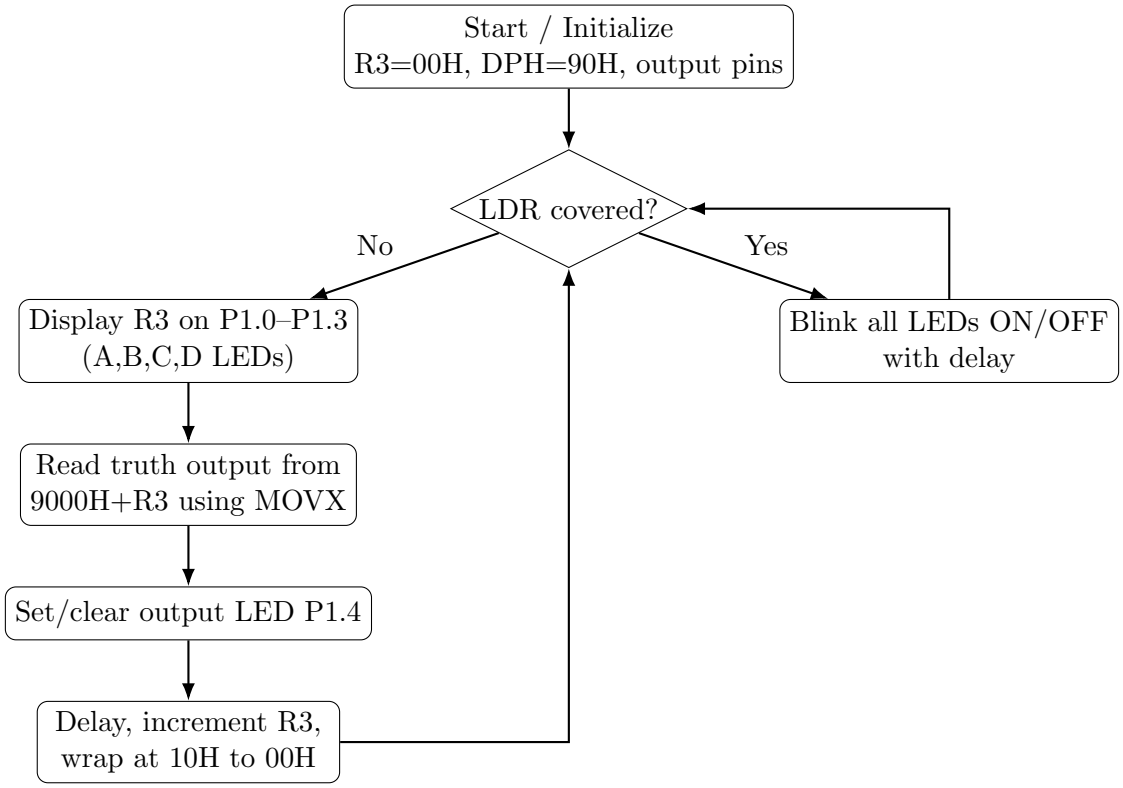
   Add LDR-based mode switching such that covering the LDR causes all five LEDs to blink; uncovering restores normal truth-table scanning from 0000.

2. Flow Chart

Part (b): Sequential 4-LED blink



Part (c) : Truth-table scan with LDR mode



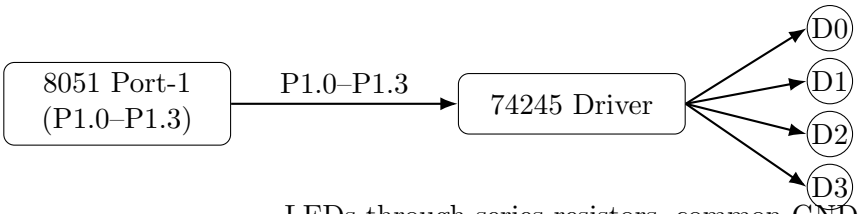
3. Circuit Diagram, Pin Description, and Components

Components used:

- 8051 microcontroller trainer kit
- Breadboard and 26-pin FRC cable
- IC 74245 (bus transceiver/driver for LED interfacing)
- 5 LEDs (4 for input bits + 1 for function output)
- Current limiting resistors (typically 220Ω to 1kΩ per LED)
- LDR and bias resistor (for digital light/dark sensing interface)
- External DC supply and jumper wires
- **Softwares Used**: Keil μVision, Win51

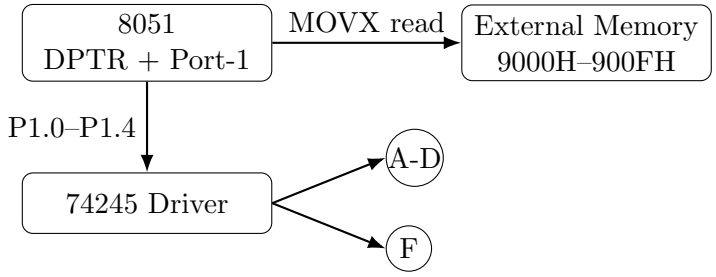
Circuit diagrams:

Part (b): Sequential 4-LED blink connection



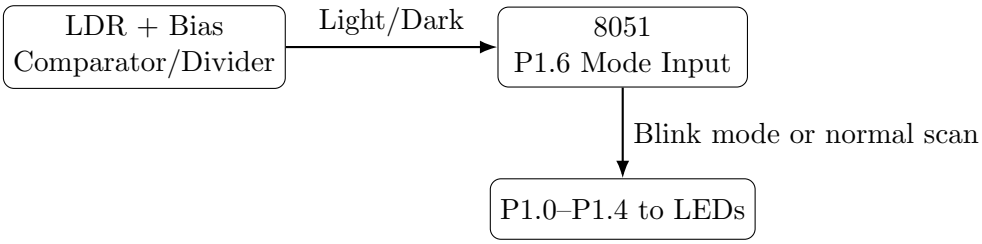
LEDs through series resistors, common GND

Part (c): Truth-table scan hardware mapping



P1.0–P1.3 show ABCD, P1.4 shows output F

Part (d): LDR-based mode switching addition



Covered LDR ⇒ blink all LEDs, uncovered ⇒ truth-table scan

Pin usage (as per program):

- P1.0 → Input LED for bit D (or mapped bit-0)
- P1.1 → Input LED for bit C (or mapped bit-1)
- P1.2 → Input LED for bit B (or mapped bit-2)
- P1.3 → Input LED for bit A (or mapped bit-3)
- P1.4 → Function output LED F(A,B,C,D)
- P1.6 → LDR status input (used in part-c)

External memory mapping:

- Truth table base address: 9000H
- Offset (0 to 15) corresponds to binary input ABCD from 0000 to 1111
- Content at each address is 00H or 01H to represent logic output

Connection details:

- 8051 Port-1 lines connected to 74245/breadboard nodes and then LEDs through series resistors.
- LDR conditioning output connected to P1.6.
- Ground of 8051 kit and external supply tied together (mandatory common reference).

4. Pseudo-code / Algorithm

Part (b): Sequential 4-LED blink

1. Clear P1.0 to P1.3.
2. Repeatedly:
   - Turn ON one LED.
   - Wait approximately 1 second.
   - Turn OFF that LED.
   - Move to next LED.

Part (c), (d): Combinational truth-table scan with LDR mode

1. Initialize index $R3 \leftarrow 00H$ and set external memory page to 90H.
2. Loop forever:
   - If LDR indicates cover condition, blink all five LEDs and reset index to 0000.
   - Else (normal mode):
     (a) Output index bits on P1.0–P1.3.
     (b) Read truth value from memory location 9000H + index.
     (c) If value is 1, set P1.4; else clear P1.4.
     (d) Delay for visibility.
     (e) Increment index; if index reaches 10H, reset to 00H.

5. Actual Code (8051 Assembly)

(a) Sequentially blink 4 LEDs

```
ORG 8100H
CLR P1.0
CLR P1.1
CLR P1.2
CLR P1.3

START:
    SETB P1.0
    ACALL DELAY
    CLR P1.0
    SETB P1.1
    ACALL DELAY
    CLR P1.1
    SETB P1.2
    ACALL DELAY
    CLR P1.2
    SETB P1.3
    ACALL DELAY
    CLR P1.3
    SJMP START

DELAY:
    MOV R2, #20
D3: MOV R1, #100
D2: MOV R0, #255
D1: DJNZ R0, D1
    DJNZ R1, D2
    DJNZ R2, D3
    RET
END
```

(b) Truth-table based combinational output

```
ORG 8100H
MOV R3, #00H ; index for ABCD = 0000 to 1111
MOV B, #01H ; compare value for TRUE output
MOV DPH, #90H ; base page for truth table (9000H)

START:
    MOV A, R3
    MOV P1, A ; show input combination on lower bits

    MOV A, R3
    MOV DPL, A ; DPTR = 9000H + R3
    MOVX A, @DPTR ; read truth table entry

    CJNE A, B, NEQ
    SETB P1.4 ; output LED ON if entry == 1
BACK:
```

```
    INC R3
    ACALL DELAY
    CJNE R3, #10H, START
    MOV R3, #00H
    SJMP START

NEQ:
    CLR P1.4 ; output LED OFF if entry != 1
    SJMP BACK

DELAY:
    MOV R2, #20
D3: MOV R1, #100
D2: MOV R0, #255
D1: DJNZ R0, D1
    DJNZ R1, D2
    DJNZ R2, D3
    RET
END
```

(c) LDR-triggered blink mode

```
ORG 8100H
MOV R3, #00H
MOV B, #01H
MOV DPH, #90H
SETB P1.6 ; configure/read LDR status line

START:
    JNB P1.6, BLINK ; if covered (low), go to blink mode

    MOV A, R3
    MOV P1, A
    MOV A, R3
    MOV DPL, A
    MOVX A, @DPTR

    CJNE A, B, NEQ
    SETB P1.4
BACK:
    INC R3
    ACALL DELAY
    CJNE R3, #10H, START
    MOV R3, #00H
    SJMP START

NEQ:
    CLR P1.4
    SJMP BACK

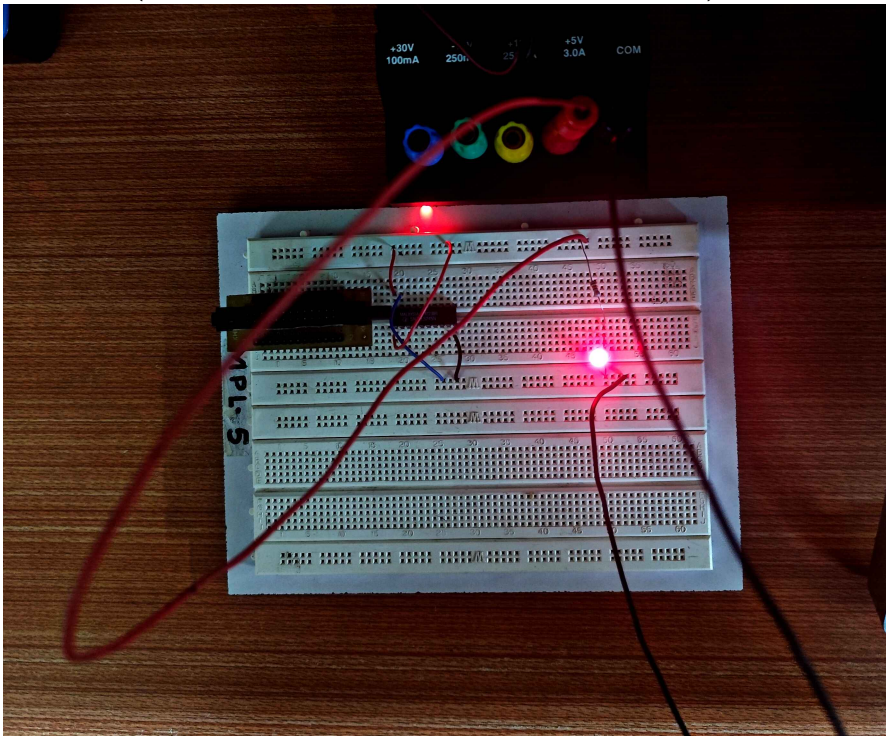
BLINK:
    MOV A, #0FFH
    MOV P1, A ; all LEDs ON
```

```
    ACALL DELAY
    MOV P1, #00H ; all LEDs OFF
    ACALL DELAY
    MOV R3, #00H
    SJMP START

DELAY:
    MOV R2, #10
D3: MOV R1, #100
D2: MOV R0, #255
D1: DJNZ R0, D1
    DJNZ R1, D2
    DJNZ R2, D3
    RET
END
```

6. Results

- Four LEDs blink sequentially with visible ON/OFF delay.
- In truth-table mode, input LEDs cycle from 0000 to 1111 and output LED follows memory content at 9000H–900FH.
- In LDR-covered condition, all five LEDs blink; when uncovered, normal scan restarts from 0000.
- **Image 1 (Testing of LED and other components):**



- **Video 1 (Sequential 4-LED blink):** Watch on YouTube

- **Video 2 (Truth-table scan 0000 to 1111):** Watch on YouTube

- **Video 3 (LDR-triggered blink mode):** Watch on YouTube

7. Discussion

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Three layers of functionality were implemented and verified in this experiment: sequential four-LED blinking, truth-table based combinational output generation, and LDR-controlled mode switching on the 8051 platform. The central learning outcome was that a combinational Boolean function can be realized efficiently through memory mapping, instead of repeatedly encoding logic terms in branching statements. By storing outputs at addresses 9000H–900FH and scanning input combinations from 0000 to 1111, a reusable firmware structure was maintained while only truth-table data needed to be modified. During hardware integration, strong dependence on proper interface discipline was observed, especially with respect to common grounding between the trainer kit and external power supply. Improved output clarity and more robust LED behavior under repeated cycling were obtained through the use of the 74245 driver. It was also verified that P1.0–P1.3 correctly represented input states, while P1.4 reflected the fetched function output, which made real-time breadboard debugging more systematic. In timing analysis, software delay loops were found to be sufficient for visual demonstration, but not ideal for calibrated one-second delays, since loop duration varies with oscillator settings and instruction overhead. Through this observation, a key embedded-systems tradeoff between implementation simplicity and timing precision was highlighted, and the preference for timer-based delays in production-oriented firmware was reinforced.For the LDR extension, reliable switching between normal scan mode and all-LED blink mode was confirmed through the P1.6 input condition. Test repeatability was improved and deterministic behavior was achieved by resetting the sequence index to 0000 after LDR-triggered events.

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- **Current Limiting and LED Safety:** In the initial phase of glowing an LED, the necessity of a series resistor was verified. LEDs are current-driven devices with low forward resistance; connecting them directly to a power supply without a current-limiting resistor risks permanent thermal damage to the diode.
- **Timing and Sequential Control:** For the sequential 4-LED blink, the use of the `ACALL` instruction to a nested `DJNZ` delay subroutine proved effective for creating a human-perceptible delay of approximately 1 second. By using `SETB` and `CLR` on specific bits of Port 1, we achieved independent control of the LEDs without affecting other pins on the same port.
- **Port Configuration and Sink/Source Issues:** A critical observation was made regarding Port 1 initialization. When moving values to the accumulator and then to the port, the LEDs initially remained in an **always-on** state regardless of the logic sent. This was resolved by explicitly configuring the port pins by writing a logic '0' to the port initially. This ensured the internal pull-ups and FETs were in a known state for the subsequent output operations.
- **Buffering with IC 74245:** Due to the limited current sourcing capability of the 8051 CMOS ports, the LEDs could not be powered directly at high brightness. The **74245 Octal Bus Transceiver** was used as a buffer/driver. By using the microcontroller signal to drive the high-impedance inputs of the 74245, the IC was able to source the necessary current from the VCC rail to the LEDs, ensuring stable operation and protecting the microcontroller from overcurrent.
- **Signal Conditioning for LDR Interfacing:** During the implementation of the LDR-triggered mode, we encountered an issue where the resistive divider output (LDR and bias resistor) only reached approximately **2.5V** in the "High" state. Since the 8051 often requires a higher voltage threshold for a reliable logic '1', the signal was not being registered.

  To solve this, we utilized the 74245 IC not just as an output driver, but as a **level regenerator**. Because the V_{IH} (Input High Voltage) threshold of the 74245 is lower (around 2.0V), it correctly interpreted our 2.5V signal as 'High' and outputted a full 5V logic signal to the 8051 Port 1.6. This effectively acted as a comparator stage, allowing the microcontroller to correctly distinguish between the covered and uncovered states of the LDR.
- **Logic Flow and Memory Access:** The use of `MOVX` to fetch truth-table values from external memory (9000H range) demonstrated the importance of the `DPTR` register in handling 16-bit addresses. The integration of the LDR check at the start of each loop ensured that the system remained responsive to environmental changes, resetting the truth-table scan to 9000H whenever the LDR was uncovered after a blink cycle.